SKILL WALLET
A SmartBridge Product

SMARTBRIDGE
Let's Bridge the Gap

Problem-Solution Fit

Date	15 July 2026
Team ID	xxxxxx
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviours, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Farmers (Primary Users) • Agricultural Officers • Agriculture Students & Researchers • Government Agricultural Departments	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** • Limited technical knowledge. • Limited internet access in rural areas. • Budget constraints. • Lack of real-time agricultural support.	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** **Pros:** Agricultural experts provide guidance and Soil testing laboratories offer reports. **Cons:** Expert consultation is not always available, Recommendations are often manual and time-consuming.

2. PROBLEMS / PAINS
+ ITS FREQUENCY **[PR]**

- Difficulty selecting the most suitable crop. (High)
- Lack of expert agricultural guidance. (High)
- Low crop productivity due to incorrect crop selection. (High)
- Financial losses caused by poor harvests. (Medium)

9. PROBLEM ROOT / CAUSE [RC]

- Lack of accurate crop recommendation systems.
- Changing climate conditions.
- Insufficient knowledge of soil nutrients.
- Manual decision-making based on experience.
- Limited access to agricultural experts.

7. BEHAVIOR
+ ITS INTENSITY **[BE]**

- Relies on traditional farming practices.
- Consults local farmers and agricultural officers.
- Uses previous farming experience.
- Searches online for farming advice.
- Makes crop decisions before each cultivation season.

Intensity: High

3. TRIGGERS TO ACT [TR]

- Beginning of a new farming season.
- Soil testing completed.
- Change in weather conditions.
- Need to maximize crop productivity and profit.

4. EMOTIONS
BEFORE / AFTER **[EM]**

Before: Confused, Uncertain, Worried and Stressed about crop failure.

After: Confident, Satisfied, Hopeful and motivated about better yield.

10. YOUR SOLUTION [SL]

OptiCrop: Smart Agricultural Production Optimization Engine is a Machine Learning-based web application that recommends the most suitable crop using soil nutrients (N, P, K) and environmental factors such as temperature, humidity, pH, and rainfall. It helps farmers make data-driven decisions, improve crop productivity, reduce financial losses, and promote sustainable agriculture.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE :

- OptiCrop Web Application
- Agricultural Websites
- Government Agriculture Portals
- YouTube Farming Channels

OFFLINE:

- Agricultural Officers
- Soil Testing Centers
- Local Farmers
- Farmer Meetings and Workshops